

Nutrition Atlas 2.0

Protein

Protein

Amino acids are building blocks of proteins which are vital for various body functions.

After consumption, protein is converted to amino acids by the body and then absorbed.

Skin, bone, muscle, hair, nails, and internal organs are largely made of protein.

Each gram of protein provides 4 Kcal.

Types of amino acids

Essential amino acids

Essential amino acids are required for the body to carry out important body functions. As these cannot be synthesized in the body, it must be obtained from the diet.

There are nine essential amino acids:

- Histidine
- Isoleucine
- Lysine
- Methionine
- Phenylalanine
- Threonine
- Valine
- Leucine
- Tyrosine

Non essential amino acids

Non essential amino acids are those that can be made in the body.

These amino acids are obtained from the food containing essential amino acids or during the breakdown of body proteins.

Complete vs. Incomplete protein: What's the difference?

Dietary proteins have different combinations of amino acids and they are categorised according to the amount of essential amino acids they contain.

Complete proteins

Proteins that adequately supply each of the nine essential amino acids are known as complete proteins.

Complete protein sources come from animal products such dairy, eggs, meat, poultry, fish, and soy.

Incomplete proteins

Proteins that lack one or more of the essential amino acids are considered incomplete proteins.

The majority of plant-based foods, including grains, beans, peas, nuts, seeds, and vegetables, are sources of incomplete protein.

Two or more incomplete protein sources can be combined to get all the nine essential amino acids.

For instance, nuts and legumes are poor in the amino acid methionine, whereas grains are low in the amino acid lysine.

When grains and legumes are consumed together (such as khichdi/Pongal, peanut butter on whole wheat bread), they form a complete protein.

What are the functions of protein?

- Protein is required to build and repair all body tissues.
- It is used for building muscles and organs.
- To make hormones and antibodies.
- To make new blood cells and necessary enzymes.
- It assists in the movement of substances in the body.

What are the good sources of proteins?

The good sources of protein include:

- Milk and milk products such as cheese, and yoghurt.
- Animal foods such as egg, meat and poultry, seafood (especially fish and shellfish).
- Plant foods such as nuts and seeds, pulses and legumes (beans, peas, and lentils), and soy products.

Animal foods are best source of proteins, however they are also high in saturated fat.

Therefore, more often choose chicken with skin removed, fish and plant proteins over meat such as beef, pork and lamb which have more saturated fat.

Some tips to make sure that the protein you consume is low in fat

- Trim or drain fat from meats before or after cooking and remove poultry skin before eating.
- Follow cooking methods that involve less fat such as baking, broiling, grilling, or steaming.

RDA and EAR for Protein for Indians - ICMR-NIN, 2020

The source presents categories for protein recommendations by life stage, including infants, children (boys and girls), men, women, and elderly. The numerical values are not legible in the extracted PDF text, so they have not been reconstructed.

Life stage	Source text
Infants	Listed in the source; numerical RDA/EAR value not

	legible in extracted text.
Children – Boys	Listed in the source; numerical RDA/EAR value not legible in extracted text.
Children – Girls	Listed in the source; numerical RDA/EAR value not legible in extracted text.
Men	Listed in the source; numerical RDA/EAR value not legible in extracted text.
Women	Listed in the source; numerical RDA/EAR value not legible in extracted text.
Elderly	Listed in the source; numerical RDA/EAR value not legible in extracted text.

What are the signs and symptoms of protein deficiency in the body?

Protein deficiency rarely occurs as an isolated condition. It usually accompanies a deficiency of dietary energy and other nutrients resulting from insufficient food intake.

The symptoms are most commonly seen in deprived children in poor countries where protein intake is exceptionally low.

Physical signs include stunting, poor musculature, edema, thin and fragile hair, and skin lesions.

Biochemical changes include low serum albumin and hormonal imbalances.

In adults the most notable signs are edema, loss of muscle mass and hair.

Rich sources of Protein

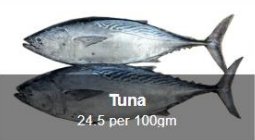
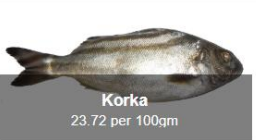
Top Sources

- Fishes And Other Sea Foods
- Meat And Poultry
- Pulses and legumes
- Oils, Fats, Nuts & oilseeds
- Vegetables
- Roots and tubers
- Green leafy vegetables
- Fruits
- Spices and Herbs
- Dairy
- Mushrooms

Rich sources of Protein

Top Sources

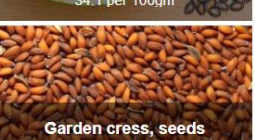
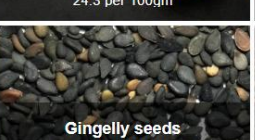
Fishes And Other Sea Foods

 Eggs, cat fish 24.68 per 100gm	 Tuna 24.5 per 100gm	 Stingray 23.98 per 100gm	 Korka 23.72 per 100gm	 Milk fish 23.66 per 100gm
 Shark, hammer head 23.4 per 100gm	 Kite fish 23.07 per 100gm	 Singhi 22.8 per 100gm	 Red snapper 22.79 per 100gm	 Silan 22.72 per 100gm

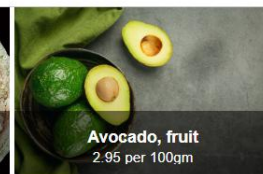
Pulses and legumes

 Soya bean, white 37.8 per 100gm	 Soya bean, brown 35.58 per 100gm	 Goa beans 32.7 per 100gm	 Khesari, dhal 28.2 per 100gm	 lentil dal 24.35 per 100gm
 Green gram, dal 23.88 per 100gm	 Black gram, dal 23.06 per 100gm	 Peas, roasted 22.9 per 100gm	 Lentil whole, yellowish 22.87 per 100gm	 Green gram, whole 22.53 per 100gm

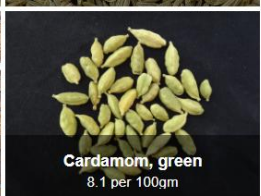
Oils, Fats, Nuts & oilseeds

 Water melon seeds (kernel) 34.1 per 100gm	 Groundnut, roasted 26.2 per 100gm	 Pumpkin seed 24.3 per 100gm	 Ground nut 23.65 per 100gm	 sunflower seeds 23.53 per 100gm
 Garden cress, seeds 23.36 per 100gm	 Pistachio nuts 23.35 per 100gm	 Gingelly seeds 21.7 per 100gm	 Mustard seeds 19.51 per 100gm	 niger seeds 18.92 per 100gm

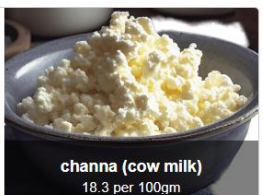
Fruits

 Black Jamun 4.8 per 100gm	 Manila tamarind 3.56 per 100gm	 Apricot, dried 3.17 per 100gm	 Wood apple 13.14 per 100gm	 Avocado, fruit 2.95 per 100gm
 Tamarind, pulp 2.92 per 100gm	 Durian 2.8 per 100gm	 Raisins, dried, golden 2.76 per 100gm.4	 Jack fruit, ripe 2.74 per 100gm	 Bael fruit 2.63 per 100gm

Spices and Herbs

 Fenugreek seeds 25.41 per 100gm	 Poppy seeds 20.31 per 100gm	 Omum 15.89 per 100gm	 Cumin seeds 13.91 per 100gm	 Chillies, red 12.69 per 100gm
 Coriander seeds 10.66 per 100gm	 Pippali 10.53 per 100gm	 Pepper, black 10.12 per 100gm.4	 Cardamom, green 8.1 per 100gm	 Turmeric powder 7.66 per 100gm

Dairy

 Skimmed milk powder (cow milk) 38 per 100gm	 whole milk powder (cow milk) 25.8 per 100gm	 Cheese 24.1 per 100gm	 Paneer (cow milk) 18.86 per 100gm	 channa (cow milk) 18.3 per 100gm
 Khoa 16.34 per 100gm	 channa (buffalo milk) 13.4 per 100gm	 milk, buffalo 3.68 per 100gm.4	 milk, goat 3.3 per 100gm	 Milk, cow 3.26 per 100gm

Mushrooms

 Oyster mushroom, dried 19.04 per 100gm	 Button mushroom, fresh 3.68 per 100gm	 Shiitake mushroom, fresh 3.19 per 100gm	 Chicken mushroom, fresh 1.84 per 100gm
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